

Calculator Test Report

Plan & forecast — 18 calculators · math, edge cases & validation · 2026-06-16

29

TESTS

29

PASSED

0

FAILED

Summary

✓ **All 29 checks pass.** Every calculator's output matched an independently hand-computed value (within tolerance), edge cases (0% rate, fully-funded, payment $\leq$ interest) behaved correctly, and all input-validation rules (age ordering, zero terms, down-payment $\geq$ price) rejected bad input as expected. Covers all 18 calculators across Retirement, Family, Save & invest, Debt & loans, Home, and Income & worth.

Results 29/29 pass

CALCULATOR	SCENARIO	INPUTS	RESULT CHECKED	STATUS
retirement	Typical (short of target)	age: 35 · retire: 65 · savings: 40000 · monthly: 600 · income: 60000 · ret: 6	projected=843612.03, target=1500000, extra=653.44, onTrack=false	PASS
retirement	On track	age: 35 · retire: 65 · savings: 40000 · monthly: 2000 · income: 40000 · ret: 6	onTrack=true	PASS
retirement	Edge: 0% return	age: 50 · retire: 60 · savings: 100000 · monthly: 1000 · income: 20000 · ret: 0	projected=220000	PASS
retirement	Validation: retire $\leq$ age	age: 65 · retire: 60 · savings: 1 · monthly: 1 · income: 1 · ret: 5	—	PASS
million	Typical	savings: 25000 · monthly: 700 · ret: 7	years=29.33	PASS
million	Validation: monthly 0	savings: 0 · monthly: 0 · ret: 7	—	PASS
college	Typical	childAge: 5 · startAge: 18 · cost: 120000 · savings: 8000 · ret: 5	need=477.83	PASS
college	Validation: start $\leq$ child	childAge: 10 · startAge: 5 · cost: 1 · savings: 1 · ret: 5	—	PASS
healthcare	Typical	age: 40 · retire: 65 · life: 88 · annual: 7000 · ret: 5	total=161000, monthly=270.36	PASS
healthcare	Validation: life $\leq$ retire	age: 40 · retire: 65 · life: 60 · annual: 7000 · ret: 5	—	PASS
emergency	Typical (underfunded)	expenses: 3500 · months: 6 · savings: 4000 · fillMonths: 12	target=21000, gap=17000, monthly=1416.67, pct=19	PASS
emergency	Fully funded	expenses: 2000 · months: 3 · savings: 8000 · fillMonths: 6	gap=0	PASS
savings-goal	Typical	goal: 20000 · savings: 2000 · years: 3 · ret: 4	need=464.77	PASS
savings-goal	Validation: years 0	goal: 1 · savings: 0 · years: 0 · ret: 4	—	PASS
invest-growth	Typical	start: 10000 · monthly: 300 · years: 20 · ret: 7	end=196665.39	PASS
mortgage	Typical 30yr	amount: 350000 · rate: 6.5 · years: 30	monthly=2212.24	PASS

CALCULATOR	SCENARIO	INPUTS	RESULT CHECKED	STATUS
mortgage	Edge: 0% rate	amount: 360000 · rate: 0 · years: 30	monthly=1000	PASS
mortgage	Validation: years 0	amount: 1 · rate: 5 · years: 0	—	PASS
auto-loan	Typical 5yr	amount: 30000 · rate: 7 · years: 5	monthly=594.04	PASS
credit-card	Typical payoff	balance: 6000 · apr: 24 · payment: 300	months=26, interest=1800	PASS
credit-card	Edge: payment ≤ interest (never)	balance: 6000 · apr: 24 · payment: 100	never=true	PASS
dti	Good (20%)	debt: 1200 · income: 6000	ratio=20, good=true	PASS
dti	High (50%)	debt: 3000 · income: 6000	ratio=50, good=false	PASS
afford-home	Typical	income: 90000 · down: 40000 · rate: 6.5 · debts: 400	price=305794.18	PASS
rent-buy	Typical	price: 350000 · down: 40000 · rate: 6.5 · rent: 2200	own=2717.74	PASS
rent-buy	Validation: down>=price	price: 100000 · down: 120000 · rate: 6.5 · rent: 1000	—	PASS
cd-savings	Typical	amount: 10000 · apy: 4.5 · years: 5	end=12461.82	PASS
take-home	Typical \$80k single	gross: 80000 · retirePct: 6 · statePct: 5	net=56695, rate=23.13	PASS
net-worth	Typical	cash: 15000 · investments: 60000 · home: 350000 · mortgage: 280000 · loans: 18000 · cards: 3000	nw=124000	PASS

GetGuac calculator suite · 29 scenarios across 18 calculators · expected values independently computed · 29 pass · 0 fail · generated 2026-06-16. Methods: future value, annuity payment, amortized loan payment, credit-card payoff (logarithmic), 28/36 affordability, single-filer federal brackets.